23/7/26

PRACTICAL - 1

AIM:

Study of various Network commands used in Linux and Windows.

BASIC NETWORKING COMMANDS:

1. arp-a:

ARP is short form of address resolution protocol, It will show the IP address of your computer along with IP address and MAC address of your router.

Output:

Interface: 192.168.56.1 --- 0xb

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.102.18	01-00-5e-7f-66-12	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 172.16.11.4 --- 0xd

Internet Address	Physical Address	Type
172.16.8.1	7c-5a-1c-cf-bc-45	dynamic
172.16.8.2	80-7c-62-71-a5-d3	dynamic
172.16.8.172	7c-57-58-35-0d-3d	dynamic
172.16.11.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.251	01-00-5e-00-00-16	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

2. hostname:

This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output:

LAP-Grace

3. ipconfig /all:

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

Output:

Windows IP configuration:

Host Name . . . . . : LAP-Grace
Primary DNS suffix . . . . . :
Node Type . . . . . : Mixed
IP routing Enabled . . . . . : No
WINS Proxy Enabled . . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS suffix . :
Description . . . . . : VirtualBox Host-only Ethernet Adapter
Physical Address . . . . . : DA-00-27-00-00-08
DHCP Enabled . . . . . : No
Auto configuration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::c259:e653(Preferred)
IPv4 Address . . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 738852903
DHCPv6 Client DUID . . . . . : 00-01-00-01-30-9A-09
NetBIOS over Tcpip . . . . . : Enabled.

4. nbtstat -a:

This command helps solve problems with NetBIOS name resolution. (Nbt stands for NetBIOS over TCP/IP).

Output:

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP).

NBTSTAT [[-a RemoteName][-A IP address][-c][-n] [-r][-R][-RR][-s][-S][interval]]

-a (adapterStatus) Lists the remote machine's name table given its name

-A (Adapter status) Lists the remote machine's name table given its IP address.

-c (cache) Lists NBT's cache of remote [machine] names and their IP addresses

-n (names) Lists local NetBIOS names.

-r (resolved) Lists names resolved by broadcast and via WINS.

-R (Reload) Purges and reloads the remote cache name table.

-S (Sessions) Lists sessions table with the destination IP addresses

-s (sessions) Lists sessions table converting destination IP addresses to computer NETBIOS names.

-RR (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

RemoteName	Remote host machine name.
IP address	Dotted decimal representation of the IP address.
interval	Redisplays selected statistics, pausing interval seconds between each display. Press ctrl+c to stop redisplaying statistics.

5. netstat: (network statistics)

netstat displays a variety of statistics about a computers active TCP/IP commands connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc. eg: netstat -r

Output:

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49670	LAP-Grace:51470	ESTABLISHED
TCP	127.0.0.1:49670	LAP-Grace:52039	ESTABLISHED
TCP	127.0.0.1:49670	LAP-Grace:55541	ESTABLISHED
TCP	127.0.0.1:51470	LAP-Grace:49670	ESTABLISHED
TCP	127.0.0.1:62654	LAP-Grace:49670	ESTABLISHED
TCP	127.1611.4:49776	lcmaa-ao-in-f14:https	ESTABLISHED

6. nslookup: (name server lookup)

nslookup is a tool used to perform DNS lookups in LINUX. It is used to displays DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes: interactive and non-interactive

eg: nslookup www.google.com

Output:

Server: Unknown

Address: 172.16.8.1

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:4826:7700::
2001:4860:482c:7700::
2001:4860:482b:7700::
2001:4860:4828:7700::
2001:4860:4827:7700::
2001:4860:482d:7700::
2001:4860:482a:7700::
2001:4860:4829:7700::
142.251.155.119
142.251.154.119
142.251.157.119
142.251.156.119
142.251.150.119
142.251.151.119
142.251.153.119
142.251.152.119

7. pathping:

Pathping is unique to Windows, and is basically a combination of the Ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output:

Usage: pathping [-g host-list] [-h maximum_hops]
[-i address] [-n] [-p period] [-q num_queries]
[-w timeout] [-4] [-6] target_name

Options:

-g host-list	Loose source route along host-list.
-h maximum-hops	Maximum number of hops to search for target.
-i address	Use the specified source address.
-n	Do not resolve addresses to hostnames.
-p period	Wait period milliseconds between pings.
-q num-queries	Number of queries per hop.
-w timeout	Wait timeout milliseconds for each reply.
-4	Force using IPv4.
-6	Force using IPv6.

8. Ping:

(Packet INetnet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet control Message Protocol) to communicate to other devices.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping www.facebook.com)

Output:

Pinging LAP-Grace [fe80::d093:8480:2114:f393%13] with 32 bytes of data:

Reply from fe80::d093:8480:2114:f393%13: time<1ms
Reply from fe80::d093:8480:2114:f393%13: time<1ms
Reply from fe80::d093:8480:2114:f393%13: time<1ms
Reply from fe80::d093:8480:2114:f393%13: time<1ms

Ping statistics for fe80::d073:8480:2114:f393%13:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

9. route:

Displays the IP routing table used by the operating system to route network traffic.

O/P:

ROUTE [-f][-p][-4|-6] command [destination]
[MASK netmask][gateway][METERIC metric]
[IF interface]

LINUX NETWORKING COMMANDS:

1. ip

Display IPaddress
configure Network interface
Add/Delete IP address
Manage routing tables.

i) Display IP Address

ip

Displays all network interfaces and their assigned IP addresses.

O/P:

Usage: ip [OPTIONS] object {COMMAND | help}
ip [-force] -batch filename

1: lo: <LOOP BACK, UP, LOWER_UP> mtu 65536
qdisc noqueue state unknown group default
qlen 1000
inet 127.0.0.1

2. enp0s3:
inet 10.0.2.15/24

ii) ip link show
Displays the status of all network interfaces

o/p:
1. lo: <LOOP BACK, UP, LOWER_UP>
2. enp0s3: <BROADCAST, MULTICAST, LOWER_UP>

iii) ip route
Displays the routing table of default gateway

o/p:
10.0.2.2 dev enpV0s3 proto dhcp src 10.2.2.15 metric 100 10.0.2.0/24 dev enP0s3 proto kernel scope link src 10.0.2.15 metric 100

2. ip address add
Assigns a new IP address to a network interfaces.

Ex: ip address add 192.168.1.254/24

o/p:
1. lo: <LOOPBACK, UP, LOWER_UP>
2. enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP>
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3.
inet 192.168.1.254/24 scope global enp0s3.

3. Delete an IP Address
Remove the IP address from the interface

Eg: ip address del 192.168.1.254/24 dev enp0s3.

O/P:

1. lo : <LOOPBACK, UP, LOWER_UP>
2. enp0s3 : <BROADCAST, MULTICAST, UP, LOWER_UP
   inet 10.0.2.15/24 brd 10.0.2.255 scope
   global dynamic enp0s3.

4. Bring Interface Online

Enables the Network interface

Eg: ip link set enp0s3 UP

O/p:

1. lo : mtu 65536 qdisc noqueue
   state UNKNOWN mode DEFAULT mode
   group default qlen 1000 link/loopback
   00:00:00:00:00:00 brd 00:00:00:00:00:0

2. ifconfig

It is a staple in many sysadmin's tool belt for configuration, configuring & troubleshooting Networks.

O/P:

Windows IP configuration

Unknown adapter McAfee VPN:

Media state . . . . . : Media disconnected

Connection-specific DNS suffix.:

Wireless LAN adapter Wi-Fi 3:

Media state . . . . . : Media disconnected

Connection-specific DNS suffix:

3. mtr

combines the functionality of the ping and traceroute commands.

a> mtr google.com

Shows you the statistics, including each hop with time & loss %.

O/p:

Host	Loss %	Snt	Last	Avg	Best
.172.16.8.1	0.0%	134	0.1	0.2	0.1
. 172.97.163.129	2.6%	134	3.9	5.2	1.1
???	2.2%	188	46.6	18.6	4.1

b> mtr -g google.com

show numeric IP address

c> mtr -b google.com

show numeric IP address & hostnames

d> mtr -c 10 google.com

Set the number of ping that you want

4. tcpdump:

This command is designed for capturing & displaying packets.

i> tcpdump -i eth0 -c 10 host 8.8.8.8

Filter our traffic comming from a specific host.

5. ping LAP-Grace

It display best way to test connectivity b/w two nodes.

Syntax: ping hostname

O/P:

Pinging LAP-Grace [fe80::d093:8480:2114:f393%13]
with 32 bytes of data:

Reply from fe80::d093:8480:2114:f393%13: time<1ms

Ping statistics for fe80::d093:8480:2114:f393%13:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

Result:

The Basic Networking of Linux commands are learned & executed successfully.